Class Project

Data Engineering and Data Governance

A growing retail company specializing in bicycle-related products relies on two independent operational systems:

1. An ERP system (Enterprise Resource Planning) that manages customer demographics, geographic information, and product category structures.
2. A CRM system (Customer Relationship Management) that manages customer profiles, product master data, and detailed sales transactions.

Both systems support day-to-day business operations, but they were not designed for analytical reporting. As a result, analytical queries are slow, complex, and require manual data extraction from multiple sources.

The company is on a budget, and they cannot afford a could based system. Instead they want to have a centralized Data Warehouse in SQL Server, so that they can boost their sales. Their initial initiative is to:

- Consolidate data from ERP and CRM systems
- Create a structured analytical data model
- Enable efficient SQL-based reporting
- Provide a reliable foundation for future BI dashboards

Since they are on a budget they want to hire a fresh data engineer! You are the chosen ones (since you decided to take this course!) as a part of the data engineering team responsible for development and implementing of this solution.

Here are their Expected Outcomes:

1. A structured and documented Data Warehouse schema
2. Integrated ERP and CRM data
3. A clean analytical layer optimized for querying
4. SQL-based insights derived from the consolidated dataset

Here are some examples of the questions that they have in mind:

- What is the total revenue?
- What is the total quantity sold?
- What is the average order value?

- What are total sales by month?
- What is the monthly revenue trend?
- Which month had the highest revenue?
- What is the average fulfillment time?
- Who are the top 10 customers by revenue?
- Revenue by country
- Revenue by gender
- Revenue by marital status
- Revenue by age group
- Do older customers spend more than younger customers?

You will work with CSV files extracted from two different operational systems. Here are the information regarding the ERP files:

1. **CUST_AZ12.csv:** This file provides basic demographic attributes for customers.

Column	Meaning
CID	Customer ID (unique customer identifier in ERP system)
BDATE	Birth Date of the customer
GEN	Gender of the customer

2. **LOC_A101.csv:** This file contains customer location information.

Column	Meaning
CID	Customer ID
CNTRY	Country where the customer is located

3. **PX_CAT_G1V2.csv:** This file provides product category information.

Column	Meaning
ID	Product Category ID
CAT	Main product category
SUBCAT	Product subcategory
MAINTENANCE	Indicates whether the product requires maintenance

Here are the information regarding the CRM Source Files:

1. **cust_info.csv:** This file contains detailed customer master data.

Column	Meaning
cst_id	Customer ID
cst_key	Customer business key
cst_firstname	Customer first name
cst_lastname	Customer last name
cst_marital_status	Marital status
cst_gndr	Gender
cst_create_date	Date when the customer was created in CRM

2. **prd_info.csv:** This file contains product master information.

Column	Meaning
prd_id	Product numeric ID
prd_key	Product business key (used in sales transactions)
prd_nm	Product name
prd_cost	Product cost
prd_line	Product line/category code
prd_start_dt	Product availability start date
prd_end_dt	Product availability end date (null = still active)

3. **sales_details.csv:** This file contains sales transaction data.

Column	Meaning
sls_ord_num	Sales order number
sls_prd_key	Product key
sls_cust_id	Customer ID
sls_order_dt	Order date
sls_ship_dt	Shipping date
sls_due_dt	Due date
sls_sales	Total sales amount
sls_quantity	Quantity sold
sls_price	Unit price